



WAKEFIELD

EXHAUST VALVE GRINDER TYPE 'SEG'

(for MAN-B&W engines) (Electric driven)

* Manual Grinding Feed

revised on Dec.20, 2004

* Power Supply : 440V, 3Ph, 60Hz

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WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' Safety Warnings



Carefully read instructions prior to Operations.



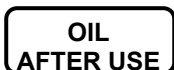
Always protect your eyes when grinding.



Use Ear-mufflers. Grinding Motor sound level 85dB at 3 meter radius.



Take care, Rotating Parts.



Clean and oil bare parts after use of the machine.

When operating the machine stay clear of the area not to be contacted by the rotating components. Always wear eye-protectors, ear-mufflers and hand protections when carrying out the grinding and lapping operations.

The electric motors should not be exposed to extreme heat or wet and humid surroundings.

Ensure the correct voltage is supplied to the motors and that cables and connections are proper.



WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' Specifications

The SEG Grinding Machine is designed for grinding of Exhaust Valve Spindles and Bottom Pieces for 2-stroke engines.

for VALVE SPINDLE STEM DIA :	30~108 mm
for BOTTOM PIECE OUTER DIA :	202~660 mm
ELECTRIC POWER:	440VAC, 3ph, 60Hz
DRIVING MOTOR:	90 W, 0-20 min ⁻¹ Reversible Direction
GRINDING MOTOR	1400 W
GRINDING SPINDLE SPEED :	4,200 min ⁻¹
GRINDING MOTOR STARTING CURRENCY:	4.5 A
MACHINE WEIGHT :	180-235 kgs
SIZE :	H; 1200-1400mm W; 1000-1200mm D; 750mm
MACHINE REST WEIGHT :	25 kgs
SIZE :	H x W x D 600 x 800 x 800mm

The Valve Spindle is secured in the machine, perfectly aligned by the Centering Chuck and Securing Chuck.

When grinding the Exhaust Bottom Piece, the Seat is centered and secured in the Chuck.

The Grinding Angle is fixed as per the Engine Makers instructions, however a Grinding Angle Adjustment Device is incorporated in the SEG machine to allow for checking and changing of grinding angle if so recommended by the Engine Makers.

The Driving Motor speed can be adjusted to slow down during the final stage of grinding operation to achieve a good finish of the gas sealing surfaces on the Valve Spindles and Valve Seats.

The cutting feed and grinding feed are manually fed.

A Roughness Template is fitted to the Angle Piece for the purpose of comparing the Valve Spindle and Seat surface with the required roughness.

Only original Wakefield parts and Grinding Wheel should be used.

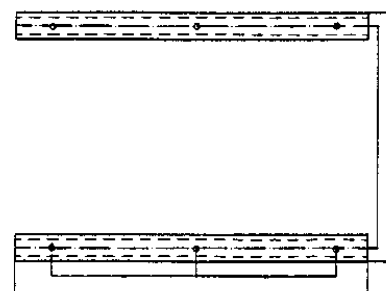
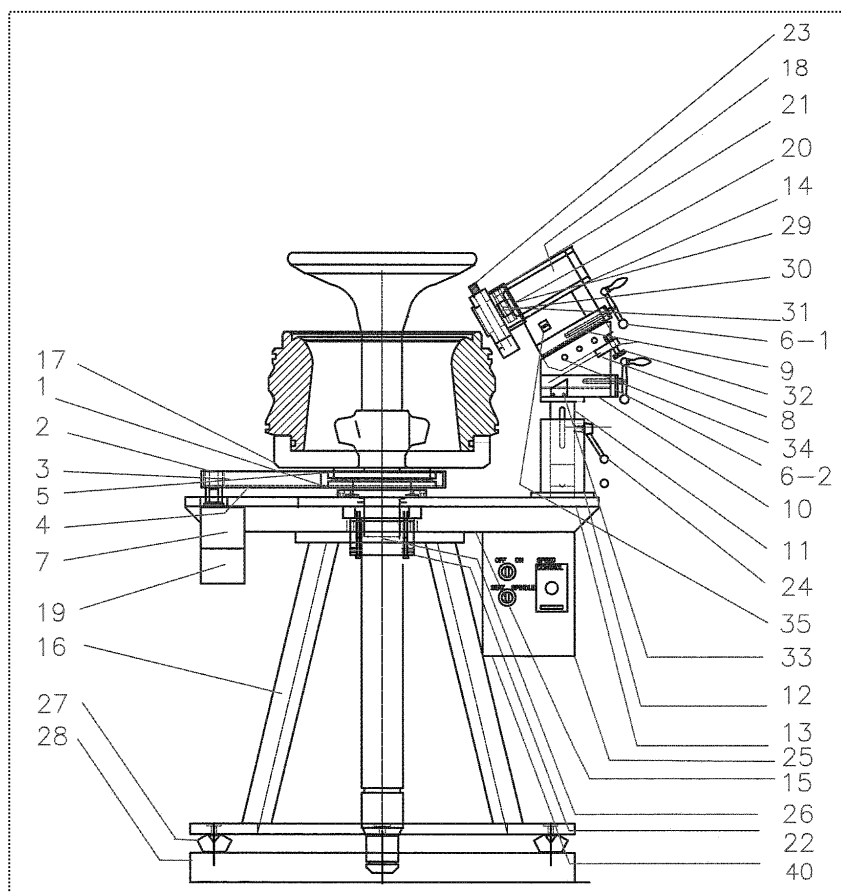
A Working Stand for dismantling of the Exhaust Valve is supplied optionally.



WAKEFIELD EXHAUST VALVE GRINDER

TYPE 'SEG'

Legend



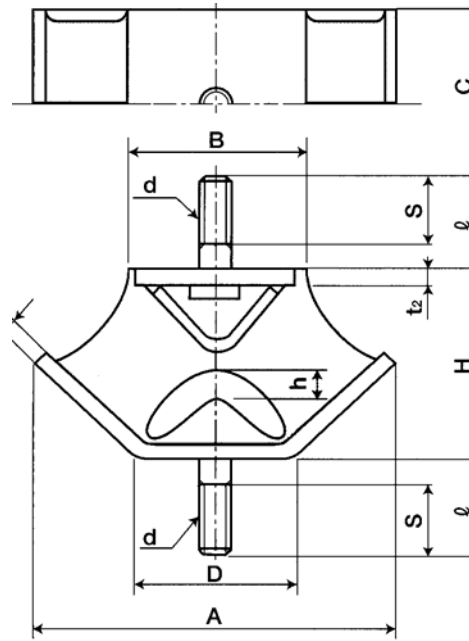
BASE FOR MACHINE
WELDED TO FLOOR
STEEL CHANNEL 75 X 75 X 9
HOLE SIZE : 12MM

1	Synchro Pulley L (Rotation)	1	19	Driving Motor (electric)	1
2	Synchro Pulley S (Rotation)	1	20	Grinding Drive Assembly	1
3	Timing Belt for Drive Gear	1	21	Grinding Drive Cover	1
4	Belt Cover	1	22	Securing Chuck	1
5	Bearing Cover	1	22-1	Securing Chuck Centering Bolts	3
6-1	Grinding Feed Handle	1	23	Grinding Wheel	1
6-2	Cutting Feed Handle	1	23-1	Grinding Wheel Cover	1
7	Gear Box	1	23-2	Dresser Assembly	1
8	Angle Feed Piece	1	23-3	Dresser	1
8-1	Securing Bolts for Angle Feed Piece	4	24	Lock Grip	1
9	Angle Feed Slide	1	25	Control Panel Box	1
10	Cutting Feed Slide	1	26	Bearing	1
11	Vertical Pipe	1	27	Shock Absorbers	4-6
12	Support	1	28	Base	2
13	Frame Plate	1	29	Synchro Pulley L (Grinding)	1
14	Grinding Spindle	1	30	Synchro Pulley S (Grinding)	1
15	Bracket	1	31	Timing Belt (Grinding)	1
16	Machine Rest	2	32	Angle Adjustment Device	1
17	Centering Chuck	1	33	Angle Indication Plate	1
17-1	Centering Chuck Knobs (for Bottom Piece)	4	34	Angle Setting Bar	1
17-2	Centering Chuck Bolts (for Valve Spindle)	4	35	Roughness Template	1
18	Grinding Motor (electric)	1	40	Cable Gland 20b	1



WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' Installation of Shock Absorbers

In order to be able to use the Grinding Machine while the ship is under way it is recommended to install the Machine with the Shock Absorbers (27) positioned as described herein.



Engine type	Shock Absorber's Parts No.	Quantity of Shock Absorbers	Dimension (mm)										
			A	B	C	D	H	h	t ₁	t ₂	d	l	S
26-35MC	EC4001	4 pcs.	63	28	30	28	35	6	4.5	3.2	8	22	17
42-60MC	EC4001	6 pcs.											
70-80MC	EC4002	4 pcs.	87	41	50	42	45	8.5	4.5	3.2	10	25	20
90-98MC	EC4002	6 pcs.											

Wakefield Exhaust Valve Grinder Type SEG has been tested suitable for use when vibrations from other machineries are present.

However, if the Grinding Machine is to be installed in a place where extremely heavy vibrations and/or shocks are present it may be required to install WAKEFIELD brand Air Spring Dampers, which are available upon request.



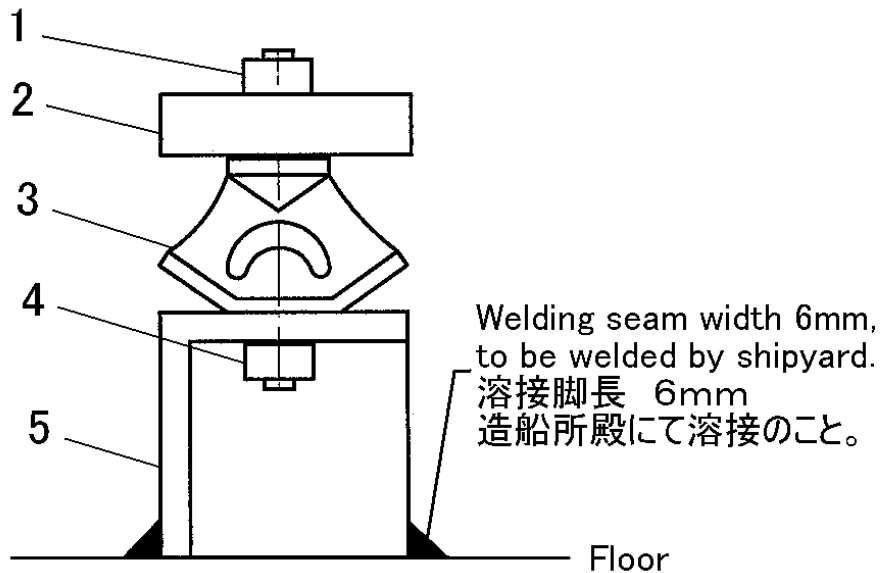
WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' Shock Absorber Wear-down

The Shock Absorber manufacturers recommend that the shock absorbers should be replaced when showing the following defects.

- 1) The rubber is found cracked more than 5mm in depth.
- 2) The rubber is detached from the metal parts.
- 3) The rubber is compressed and inflated more than 30%.
- 4) The rubber has become moist and swollen due to contact with oil.



WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' Installation of Shock Absorbers



The Base to be welded to Floor.
ベースは床に溶接のこと。

Engine types		26MC-60MC	70MC-98MC
1	Nut & Spring Washer	M8	M10
2	Machine Rest		
3	Shock Absorber	Bridgestone EC4001	Bridgestone EC4002
4	Nut & Spring Washer	M8	M10
5	Base	Steel Angle 75 x 75 x 9mm Welded to floor	

The above 1-5, to be supplied by Wakefield Corporation.

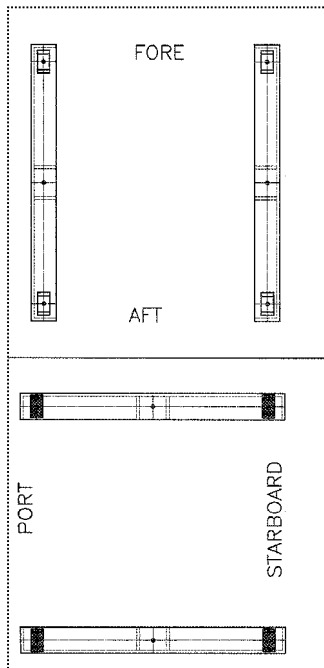


WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' Positioning of Shock Absorbers

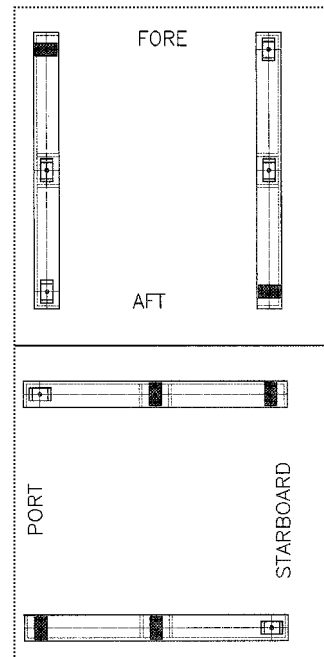
At the time of delivery from the factory, the Grinding Machine has been assembled with the Shock Absorbers positioned for installation of the machine FORE – AFT DIRECTION.

The Shock Absorber direction can be changed by loosening the nuts, both upper and lower end, and turning the Absorbers 90 degrees, then re-tightening the nuts.

**For 26,35 MC engines
For 70,80 MC/MC-C engines**



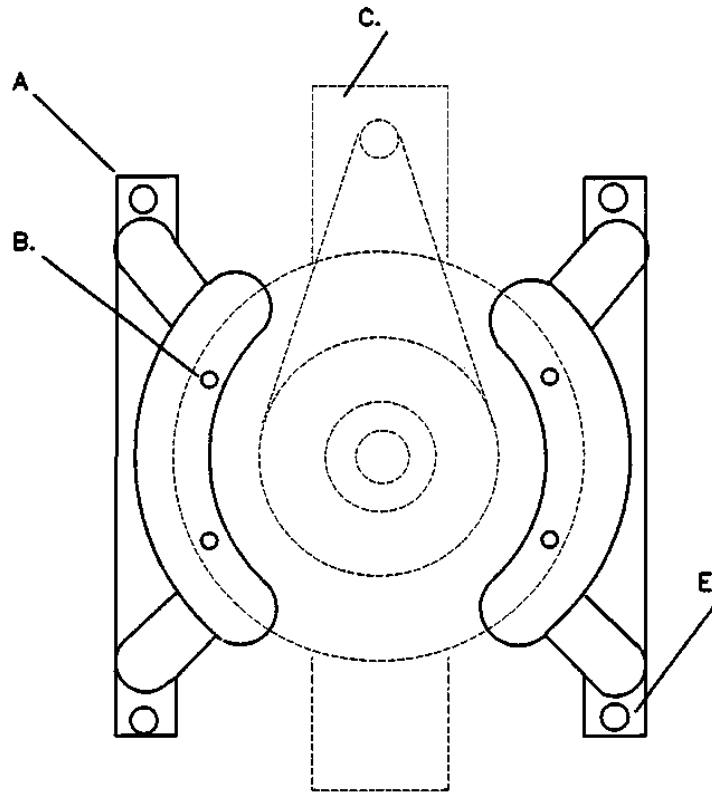
**For 42,46,50 and 60 MC/MC-C engines
For 90, 98MC/MC-C engines**





WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG'

Installation – Grinder / Machine Rest

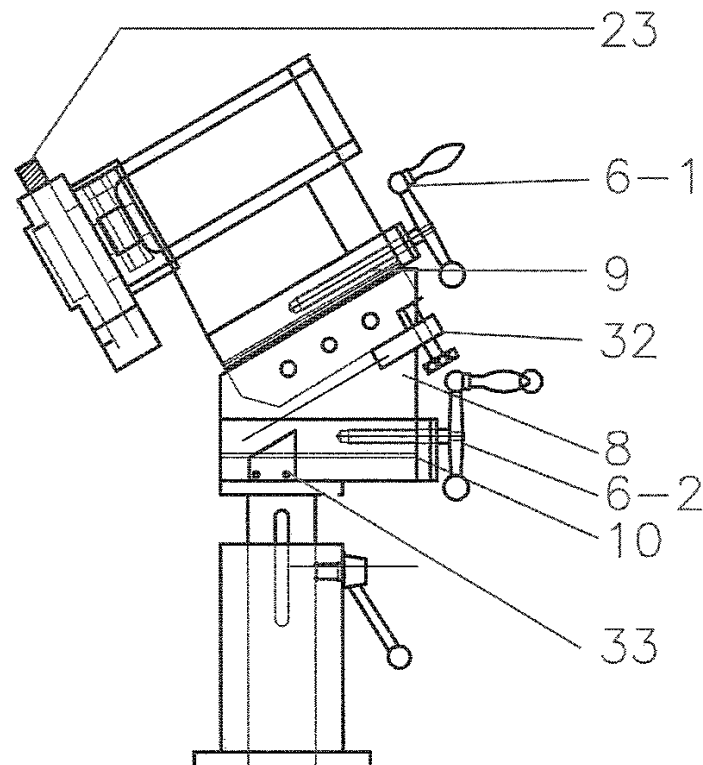


A	Machine Rest
B	Securing Bolts between Grinder and Machine Rest
C	Grinder
E	Machine Rest foundation bolts (4 or 6 pieces)



WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' GRINDING FEED

The feed of the Grinding Spindle and Grinding Wheel (23) over the seat can be achieved manually using the Manual Feed Wheel (6-1) fitted to the Angle Feed Slide Bolt.



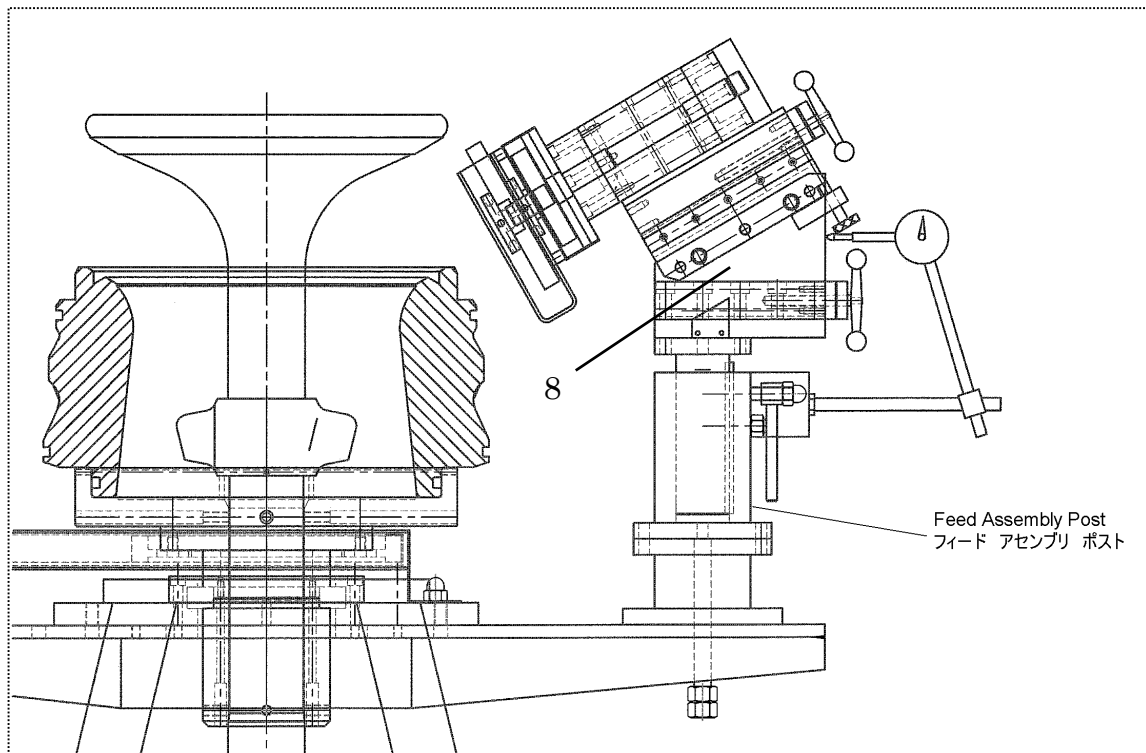
6-1	Grinding Feed Handle
6-2	Cutting Feed Handle
8	Angle Feed Piece
9	Angle Feed Slide
10	Cutting Feed Slide
23	Grinding Wheel
32	Angle Adjustment Device
33	Angle Indication Plate



WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' RECORDING CUTTING FEED

In order to monitor each increase of the cutting feed, fit a Dial Gauge magnet base on the Feed Assembly Post with the Dial Gauge Feeler on the Angle Feed Piece (8).

Each increase of the cutting feed can then be recorded by the Dial Gauge readings.





WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' PREPARATIONS & PRECAUTIONS BEFORE/WHEN COMMENCING GRINDING OPERATION

The following points should always be checked before grinding operation.
As a long period may have elapsed since the Grinding Machine was delivered from Wakefield until used for the first time.

Precautions before commencing of grinding operation

1. All rust prevention coating should be washed off and removed from moving parts and from the Chuck surfaces.
2. All Shaft Gears should be well greased and oiled.
3. All Shaft Gears should move smoothly when manually operated.
4. Ensure that the correct voltage for the electric power is supplied to the Grinding and Driving Motors.
Check that the electric cables are properly connected.
5. Check that the grinding wheel is tightly secured in the grinding wheel hub.
6. Timing Belt tension to be checked.

Precautions when commencing the grinding operation

1. Check that the correct power source is applied, condition of electric wiring and switches.
2. Check the Grinding and Driving Motors revolution.
3. Check that the Timing-Belts are properly stretched to ensure a smooth rotation speed of driving and grinding gears.
4. Ensure the Grinding Wheel is properly dressed. See Dressing.

If the vessel is rolling or listing heavily, the grinding work should be suspended.

The operator should always wear protective clothing and eye-goggles when grinding work carried out.



WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' GRINDING OF VALVE SPINDLE (1)

1. Loosen the four piece Securing Bolts (8-1) for the Angle Feed Piece(8) and position the Angle Feed Slide (9) for spindle grinding and insert Spindle Grinding Angle Setting Bar (34) into the **SPINDLE** hole and firmly tighten the four Securing Bolts(8-1) for Angle Feed Piece (two on each side).
2. After having withdrawn the Valve Spindle from the Exhaust Valve Cage, insert the Valve Spindle vertically in the Grinding Machine. Then prepare the Grinding machine for operation.
3. Lift or lower the Feed Assembly and Cutting Feed Vertical Pipe (11) to select the suitable position. Use the inserted 16mm stud underneath the post to adjust the height.

Ensure the Valve Spindle seat is in height with the Grinding Wheel (23). Secure the Lock Grip (24) firmly.

Temporarily tighten the Lower Securing Chuck Centering Bolts (22-1) and Centering Chuck Bolts (17-2) slightly by using the supplied Allen wrenches.

4. Set the Rotating Change-over Switch on the Control Panel (25) to position **SPINDLE**.

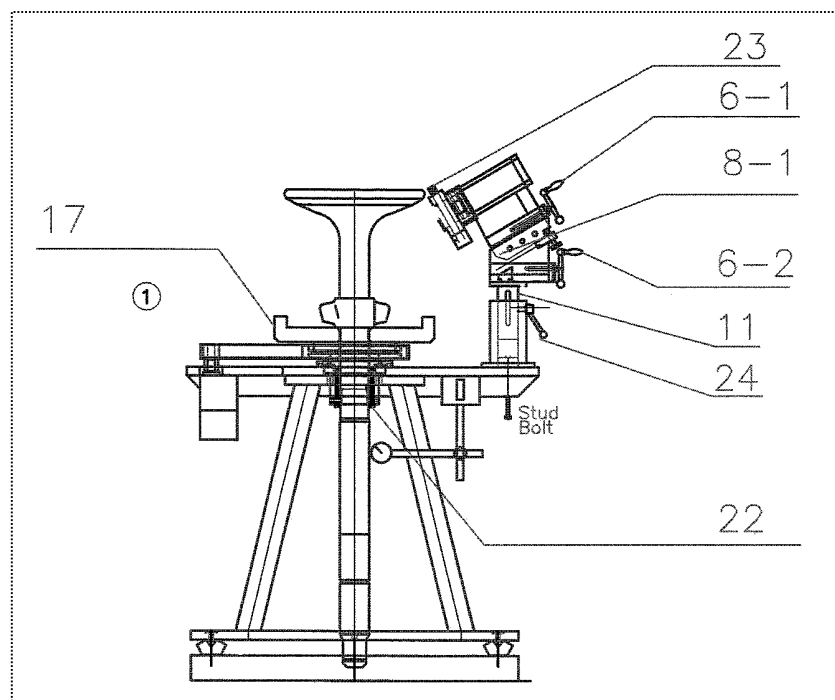
In order to achieve a perfect grinding result the rotation direction of Grinding Wheel and Spindle or Bottom Piece should be opposite where the contact is made.

5. The Valve Spindle is rotated by moving the Switch for Rotation from ST.BY to RUN, and adjusting the speed with the Speed Controller situated on the Control Panel Box.
6. Centering of the Valve Spindle is achieved as follows.

Tighten the Securing Chuck Bolts (22-1) and fit a Dial Gauge onto the Machine Bracket (15) with its feeler contacting the Valve Stem below the Securing Chuck (22).

Position ①.

Adjust to zero by manipulating the Securing Chuck Centering Bolts (22-1, 3 pieces) on the Securing Chuck (22).



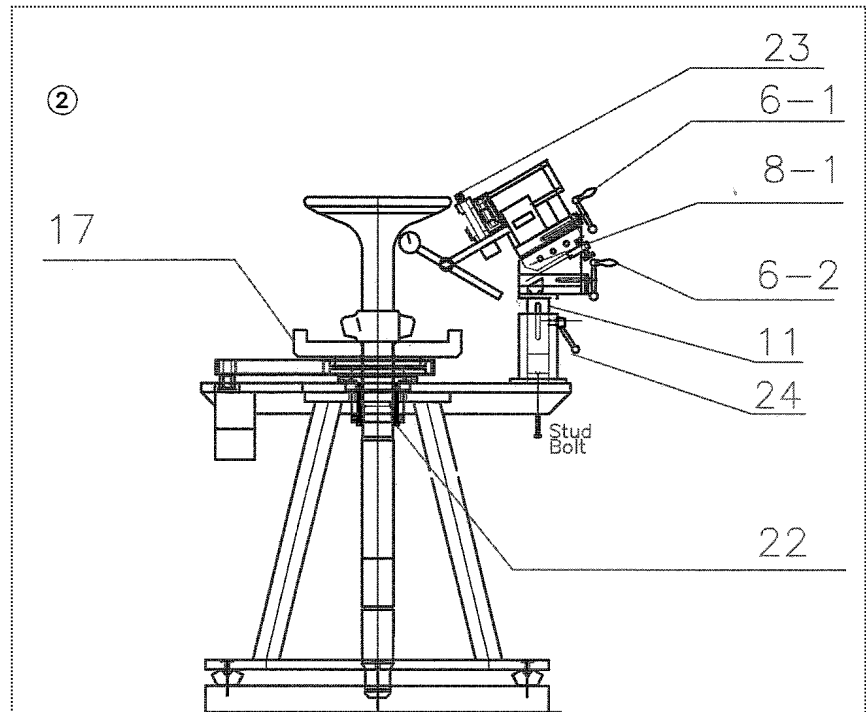
WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' GRINDING OF VALVE SPINDLE (2)

7. Secondly fit the Dial Gauge magnet base onto the Angle Feed Piece (8) with its feeler on the upper portion of the Valve Stem and adjust to zero by manipulating the Centering Chuck Bolts (17-2).

Position ②.

Rotate the machine slowly and adjust the centering as required. Tightening the Centering Bolts at the maximum plus side will give a minus reading on the dial gauge.

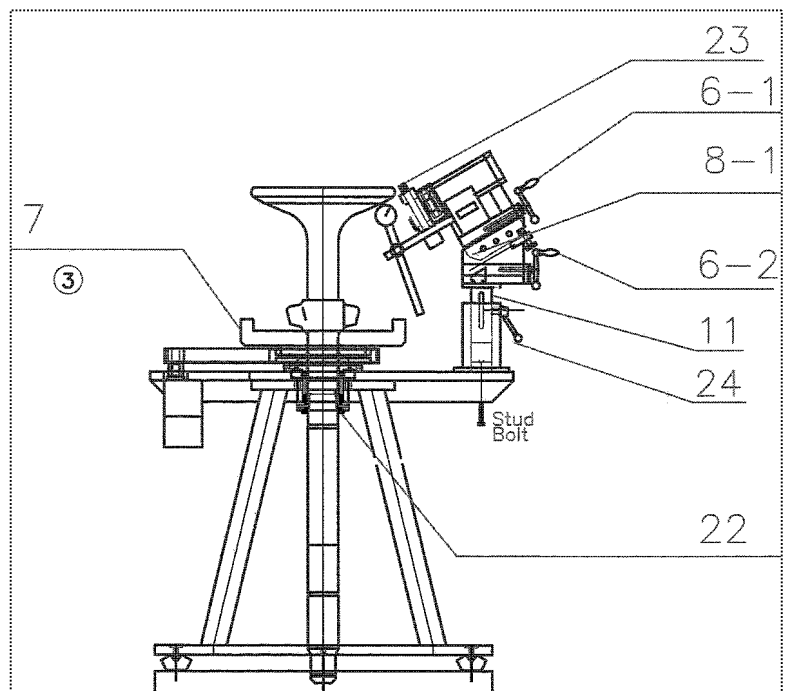
If it is too hard to tighten one Centering Bolt, slightly loosen the Centering Bolt on the opposite side and re-tighten accordingly.



8. The Valve Spindle is now basically centered, however in order to ensure a perfect centering, fit the Dial Gauge onto the Angle Piece (8) with its Feeler on the inner most part (2-4mm where the seat surface is intact) of the Valve Spindle Seat. **Position ③.**

If the deviation does not exceed 0.3mm it can be rectified by slightly loosening the Lower Securing Chuck Centering Bolts (22-1), adjusting the centering with the four Centering Chuck Bolts (17-2) and then retightening the Lower Securing Chuck Centering Bolts (22-1).

If the deviation exceeds 0.3 mm, re-check the centering at positions 1 and 2 and re-adjust accordingly. If the Spindle is found to be bent more than 0.3 mm, then consult with the engine maker.





WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' GRINDING OF VALVE SPINDLE (3)

9. The Grinding Angle of the machine is fixed at the angles recommended by the engine makers, thus no final angle setting is required. If it is necessary to change the angle see Grinding Angle Adjustment Device.

A Grinding Angle Information Plate is fixed onto Main Body.
Ex.

MAN-B&W 60MC Engine

Grinding Angle Seat: $30.0^{+0.0}_{-0.1}$

Grinding Angle Spindle: $30.3^{+0.1}_{-0.0}$

10. Commence the grinding operation by positioning the Grinding Wheel (23) at the center of the seat to be ground. The Grinding Motor and Wheel is moved into position manually rotating the Grinding Feed Slide and Cutting Feed Slide with the Manual Feed Handles (6-1 and 6-2).

11. Ensure the Grinding Wheel (23) is properly dressed. See Dressing.

12. Start the rotation and allow the Spindle to rotate slowly. Feed in the Cutting Feed Slide (10) using the Cutting Feed Handle (6-2) until the Grinding Wheel (23) slightly touches the Valve Spindle Seat.

In order to check how much material had been ground off, the Template Gauge supplied by the Engine Maker should be used.

Ensure the Rotation Change-over Switch is set to position **SPINDLE**.

13. Start the operation by moving the switch on the Speed Controller from ST.BY(stand by) to RUN and adjust the rotation speed with the Knob on the Speed Controller. As guidance it is recommended to keep the speed at 80% when grinding and approximately 60% when polishing. Confirm that the Grinding Wheel is touching the seat surface evenly.

14. Adjust the rotation speed in order to obtain the best grinding result.

15. Start the Grinding Motor by turning the Switch on the Grinding Motor. Feed slowly with the Cutting Feed Handle (6-2).

It may be necessary to increase or decrease the rotating speed to get the best grinding result.

16. The Valve Spindle will now rotate at a slow speed and the Grinding Wheel will travel from the center of the seat to the inner or outer edge of the Seat. Feed with the Grinding Feed Slide Handle (6-1). One cut will take about 10 minutes when the grinding wheel travels over the whole width of the seat.



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TYPE 'SEG'

GRINDING OF VALVE SPINDLE (4)

17. When the Grinding Wheel has reached OUTSIDE the inner or outer edge of the seat, stop the Manual Feed and change direction of the feed. The grinding wheel should be allowed to travel outside the seat in order to avoid occurrence of an edge at the ends of the seat.

Adjust the Cutting Feed slightly by manipulating the Cutting Feed Handle (6-2).

It is recommended to increase the feed to 0.01 to 0.03 mm each time, depending on grinding results. Increase the cutting feed when the Grinding Wheel is in full contact with the Valve Spindle seat. This in order to avoid excessive wear of Grinding Wheel.

18. Repeat the grinding operation until all deformation removed from the Valve Spindle seat. Finally check the correctness of grinding angle and ground off quantity using the Template Gauge supplied by the Engine makers. (See Engine Maker's Instructions) This in order to verify how much material had been ground off and if the Valve Spindle can be further used.

The correctness of Grinding Angle and the Total Run Out can also be verified as per our instructions in this manual. See "How to verify Grinding Angle upon completion of grinding" and "How to verify Total Run-out".

19. Final polishing should be effected with 0.01mm or no additional feed and with the Valve Spindle rotating slowly. The speed may need to be adjusted to get the best grinding result and obtain the surface roughness as recommended by the engine makers.

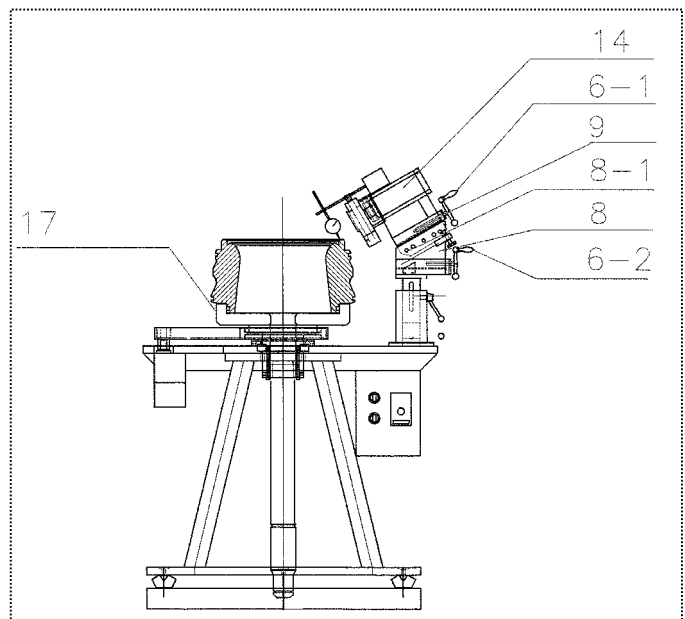
Stop the rotation by moving the switch on the Speed Controller from RUN to ST.BY.

20. After completion of grinding, hook up the Valve Spindle using the lifting tool supplied by EngineMaker, loosen the Securing Chuck Centering Bolts (22-1) and the Centering Chuck Bolts (17-2) and lift the spindle out from the Grinding Machine.



WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' GRINDING OF BOTTOM PIECE (1)

1. Remove the Bottom Piece from the Exhaust Valve Cage and place it in the Grinding Machine Centering Chuck (17).
2. Remove the Grinding Wheel cover, rotate it 180 degrees and refit it. Adjust the height of the Angle Feed Piece (8) so that the Grinding Wheel (23) is in height with the Bottom Piece seat surface.
3. Loosen the four Securing Bolts (8-1) for the Angle Feed Piece (8) and move the Angle Feed Slide (9) to allow for the Angle Setting Bar (34) to be inserted in the hole for SEAT. Then retighten the four Securing Bolts (8-1) for the Angle Feed Piece (8).
4. Fix a Dial Gauge magnet base onto the Grinding Spindle (14) with the Feeler on the outer Bottom Piece surface and center/align the Bottom Piece by manipulating the Centering Knobs (17-1) on the Centering Chuck (17).



5. Position the Grinding Wheel (23) at the center of the seat to be ground. The Grinding Motor is moved manually with the Cutting Feed Handle and Grinding Feed Handle (6-2 and 6-1).
6. Dress the Grinding Wheel. Movement of the Grinding Motor/Wheel is now achieved by manipulating the Grinding Feed Handle (6-1).
7. Feed out the Cutting Feed (6-2) until the Grinding Wheel (23) slightly touches the Bottom Piece in a position at the center of the seat.
8. Move the Rotation Change-over Switch to the **SEAT** position, switch on to RUN and adjust the rotation speed by manipulating the Speed Controller Knob for Driving Motor.
9. Rotate the Bottom Piece slowly and ensure the Grinding Wheel touches the Bottom Piece surface evenly.
10. Start the Grinding Motor by turning on the switch for Grinding Motor.
11. Carry out the grinding operations as per Grinding of Spindle.



WAKEFIELD EXHAUST VALVE GRINDER

TYPE 'SEG '

GRINDING OF BOTTOM PIECE (2)

12. Repeat the grinding operation until all deformation removed from the Bottom Piece seat. Finally check the correctness of grinding, s.a. grinding angle and how much material had been ground off, by using the Template Gauge supplied by the Engine Maker (see Engine Maker's Instructions). This in order to verify if the seat can be further used.

The correctness of Grinding Angle and the Total Run Out can also be verified as per our instructions in this manual. See "How to verify Grinding Angle upon completion of grinding" and "How to verify Total Run-out ".

13. Final polishing should be effected with 0.01 mm or no additional feed and with the Bottom Piece rotating slowly. The rotation speed may need to be adjusted to get the best grinding result and obtain the surface roughness as recommended by the engine makers.

Stop the rotation by moving the switch on the Speed Controller from RUN to ST.BY.

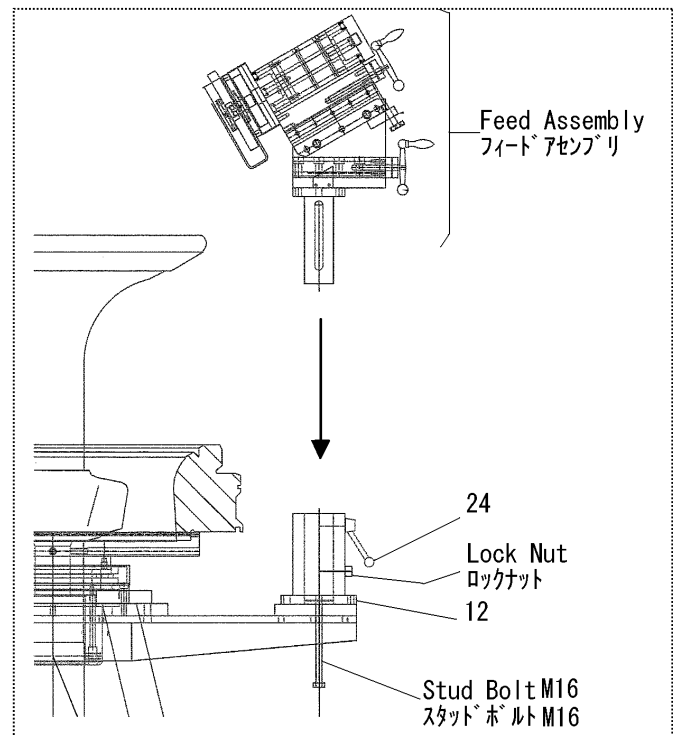
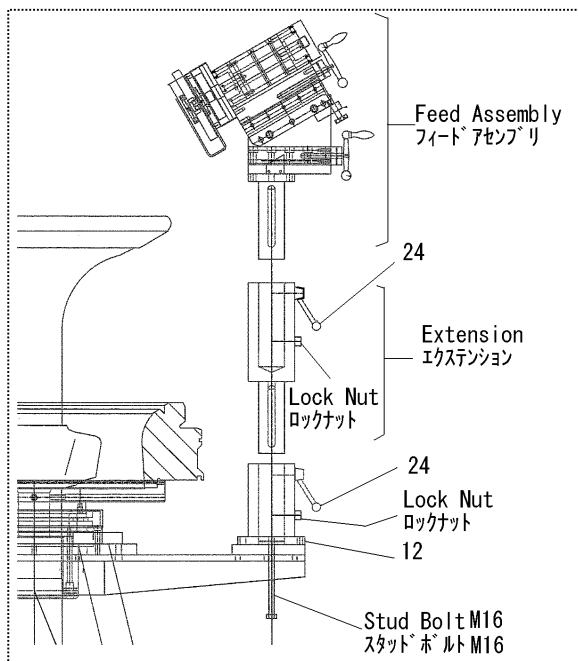
14. After completion of grinding operation, hook up the Bottom Piece using the lifting tool supplied by the Engine Maker. Loosen the Centering Chuck Knobs (17-1) and lift off the Bottom Piece from the Centering Chuck (17).



WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' GRINDING OF BOTTOM PIECE, 98MC/MC-C/ME engines

When grinding the Bottom Piece for 98MC, MC-C, ME engines, the Feed Assembly Vertical Support extension is to be removed.

- 1) Loosen the lock nuts and the lock grips (24), and screw down the M16 stud fitted under the Support (12).
- 2) Dismantle the Vertical Support extension and fit the Feed Assembly directly into the Support.
- 3) Secure it with the lock nut and the lock grip (24).

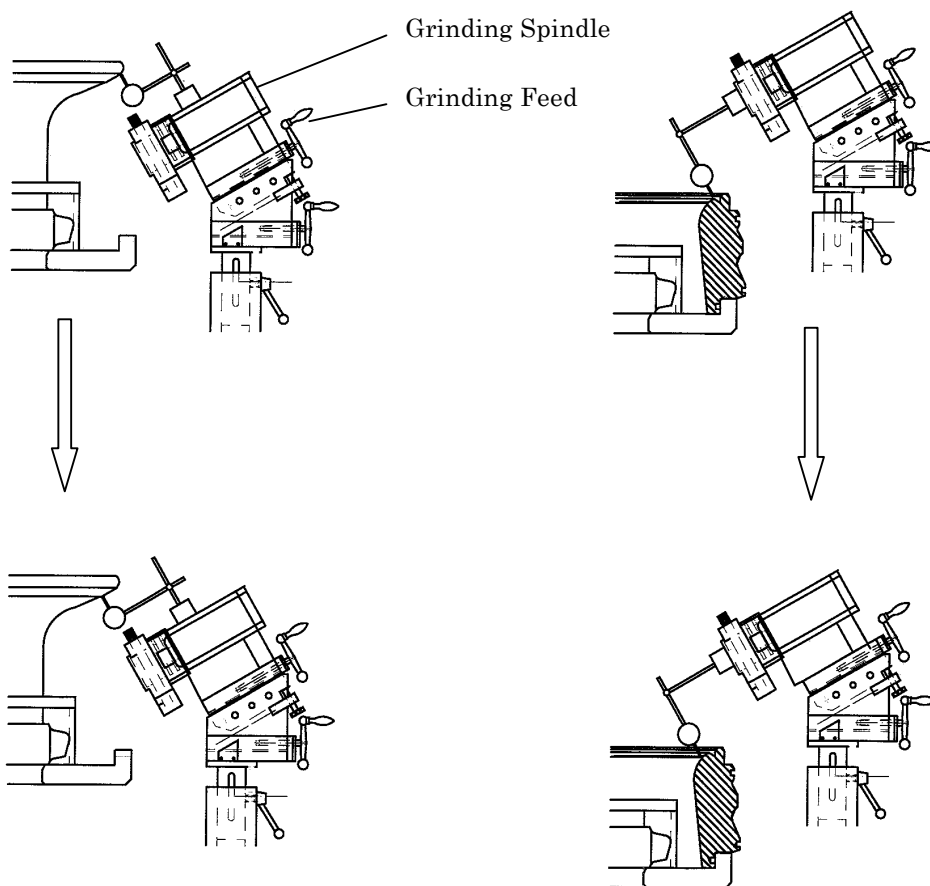




WAKEFIELD EXHAUST VALVE GRINDER **TYPE 'SEG'** **HOW TO VERIFY GRINDING ANGLE** **UPON COMPLETION OF GRINDING**

Using the Dial Gauge, fixed with its Magnetic Base to the Grinding Spindle (it may be convenient to remove the Grinding Motor), and its Feeler running on the ground Seat, the **ANGLE** of the ground seat can be verified.

Set the Dial Gauge to "0" when the Feeler is positioned at the outer end of the seat, manipulate the Grinding Feed (6-1) manually until the Dial Gauge feeler is positioned towards the inner end of the seat. Read off the deviation over a length=25 mm using a ruler or steel scale. Mark the starting and finishing points, the distance of 25mm, with a pen on the seat surface.



For the SPINDLE the angle deviation may not exceed + 0.1 degree and for the Bottom Piece maximum – 0.1 degree. Using the below described formula, the deviation can be calculated.

Dial gauge reading (on the seat width 25mm)	Deviation degree
0.131 mm	0.3
0.087 mm	0.2
0.044 mm	0.1
0.022 mm	0.05

If the deviation exceeds the stated values, one or two final grinding cuts may be required to rectify the angle of the ground seat.

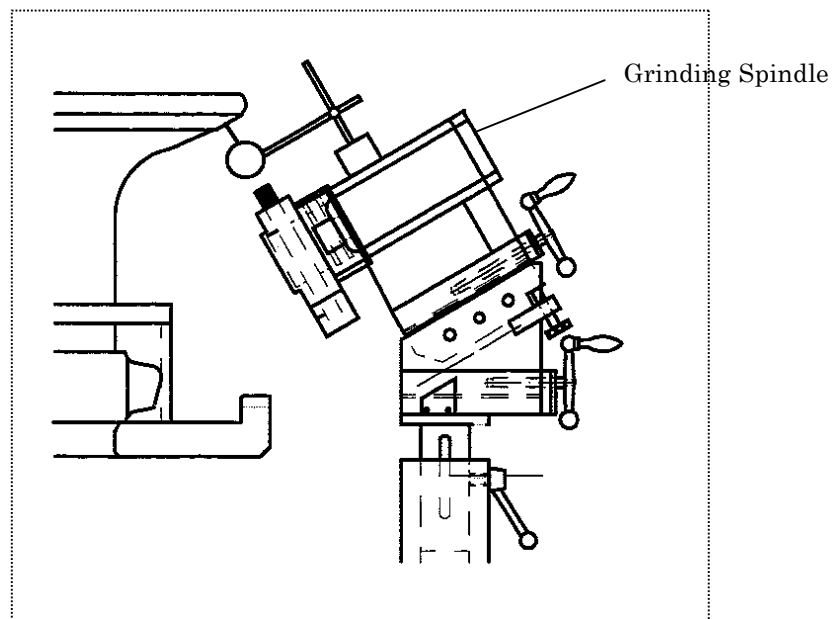


WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' HOW TO VERIFY TOTAL RUN-OUT & SURFACE ROUGHNESS

TOTAL RUN-OUT

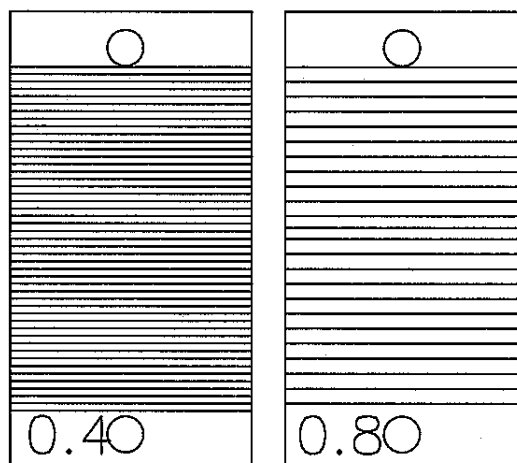
After completion of the grinding the **TOTAL RUN-OUT** can be checked by attaching the Magnet Base to the Grinding Spindle with its feeler at the center of the seat width.

Rotate the SPINDLE or BOTTOM PIECE slowly one complete turn and read off the Dial Gauge. The total deviation, between maximum and minimum, may not exceed 0.03 mm. Should the reading be more it is recommended to carry out a few more grinding/polishing cuts to bring the deviation within allowed values.



ROUGHNESS (Ra)

The surface **ROUGHNESS (Ra)** of the ground seat can be verified by touching the seat and comparing with the surface on the Roughness Template (35) fitted to the Angle Feed Piece (8) on the Grinding Machine. The roughness should not exceed $Ra\ 0.8\ \mu m$. If the surface is too rough further polishing of the seat surface may be required.

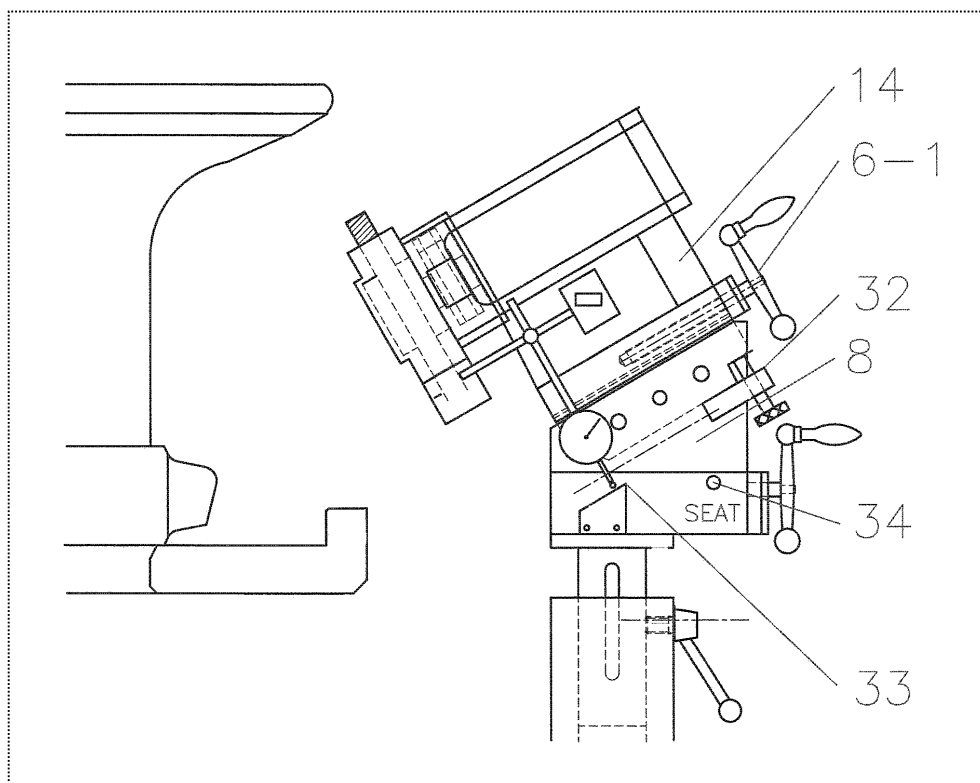




WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' HOW TO VERIFY 30.0 DEGREE GRINDING ANGLE

After having used the machine for some time the tapered holes for SEAT and SPINDLE Setting Bars may become worn or deformed, allowing the angles to change.

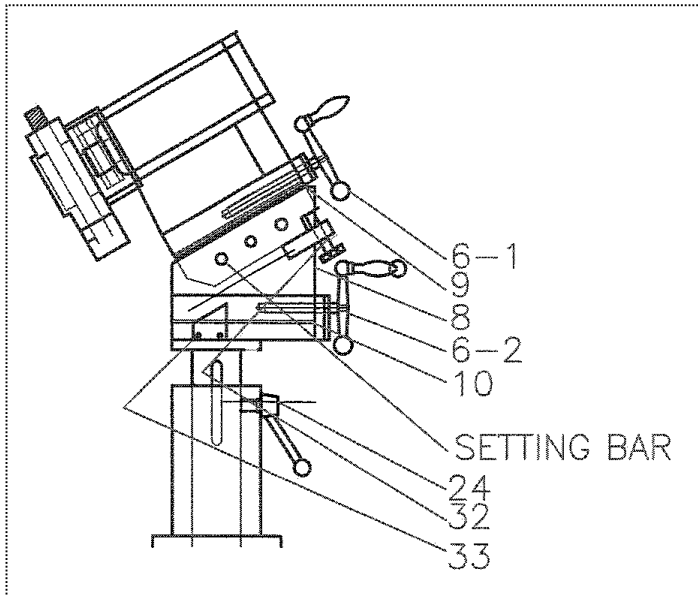
To verify that the Grinding Feed (6-1) follows a 30.0 degree line, fix a Dial Gauge Magnet Base to the Grinding Spindle (14) when the Angle Setting Bar (34) is in the SEAT position. The Feeler running on the Angle Indication Plate (33), when feeding the Grinding Feed (6-1), should give a "0" reading from end to end on the Angle Indication Plate (33).



If angle incorrect, the deviation should be corrected with the Angle Adjustment Device (32) or the tapered holes for SEAT and SPINDLE re-reamed using a 10 mm diameter tapered reamer. The holes to be reamed until deformation of the holes are removed. If the holes become too big after reaming, the standard Taper Pin to be replaced with an over-size Angle Setting Bar, e.g. 12 mm diameter Taper Pin.



WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' GRINDING ANGLE AJUSTMENT DEVICE (1)

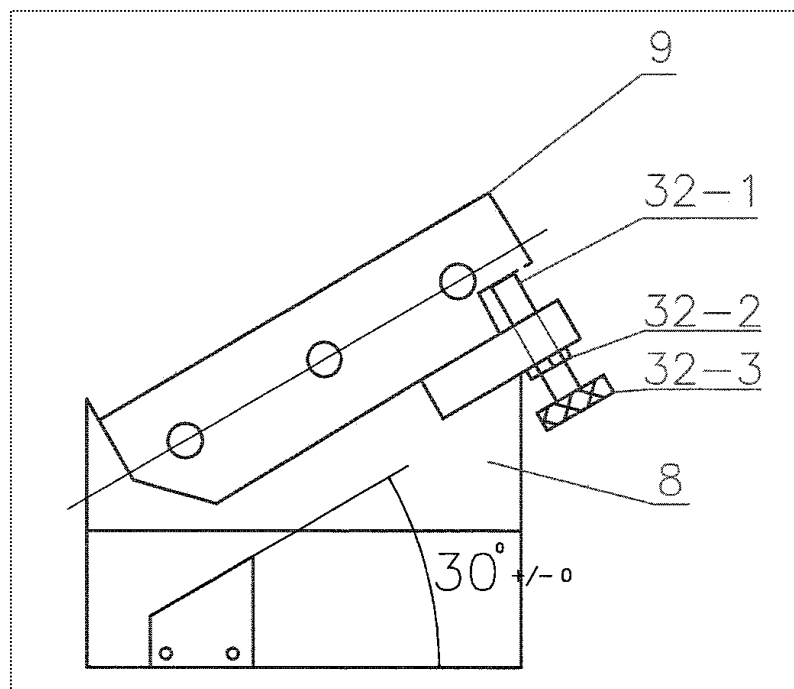


The Angle Feed consists of an Angle Feed Piece (8) and an Angle Feed Slide (9), connected at center with two Straight Dowel Pins one on each side.

This arrangement allows the Angle Piece to "swing" around the Center position as much as the distance between Angle Feed Slide underside and Angle Feed Piece allows.

Independently, a Grinding Angle Adjustment Device (32) is installed.

The device consists of a free-Angle-type Clamping Bolt (32-1), Locknut (32-2), Angle Index Knob (32-3) and Point Arrow (32-4).

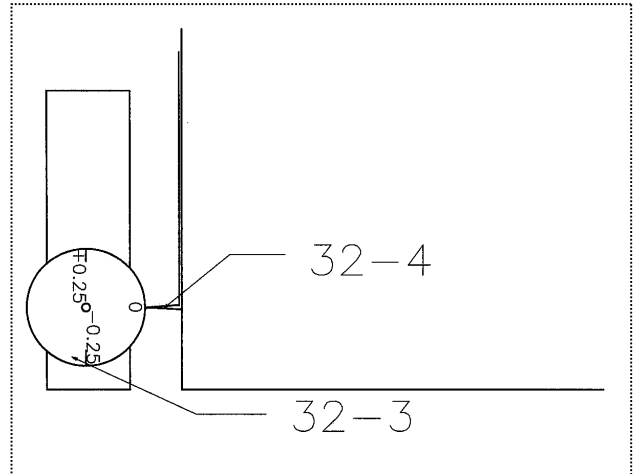




WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' GRINDING ANGLE AJUSTMENT DEVICE (2)

When the 30 degree angle has been set in a new Grinding Machine, the Grinding Angle Adjustment Device (32) is screwed into a threaded hole in the Base until the hardened Free-Angle type Clamping Bolt's (32-1) surface contacts the underside of the Angle Feed Slide (9). The Angle Index Knob (32-3) is then set so that the 0 degree mark is in line with the Point Arrow (32-4) fitted to the Base. The 0 degree mark corresponds with a 30 degree angle.

Then the Locknut (32-2) is firmly tightened.



The Knob is now secured and cannot be turned unless the Locknut is loosened.

The Locknut should remain secured with the Knob at 0 degree.

A Caution Plate stating this is fitted to the Angle Assembly side plate.

ANGLE ADJUSTMENT DEVICE

The Lock Nut should remain secured with the Knob at 0 degree.

For adjustment see Instruction Manuals.

One division on the knob=0.1 degree

If the angle needs to be changed,

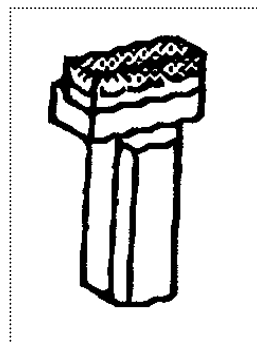
- 1) Loosen the four bolts (8-1) securing the Angle Feed Slide (9) to the Angle Feed Piece (8) and the Angle Adjustment Device locknut (32-2).
- 2) Turning the knob (32-3) clockwise will increase the angle 0.1 degree for each notch. An anticlockwise turn will decrease the angle 0.1 degree for each notch.



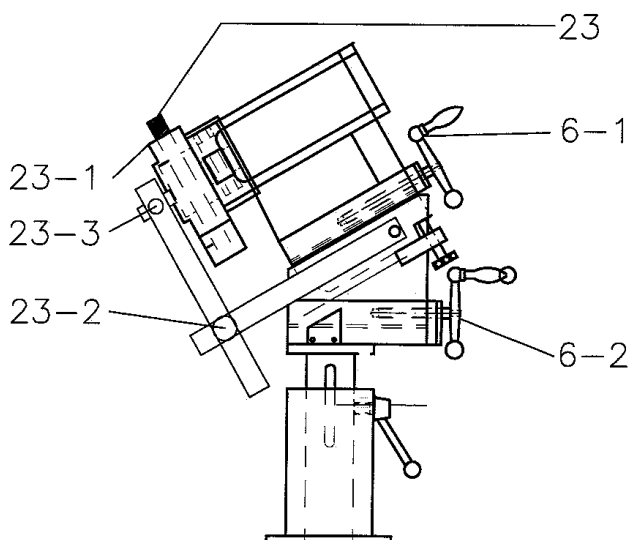
WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' DRESSING

Dressing of the Grinding Wheel is effected by using the supplied Grinding Wheel Dresser (23-3).

dresser



1. Position the Dresser Assembly Bar (23-2) so that the Diamond Dresser (23-3) is in line with the Grinding Wheel (23). Position the Diamond Dresser to contact the Grinding Wheel.
2. Start the Grinding Motor (18) and move the Grinding Wheel back and forth over the Dresser by manipulating the Manual Grinding Feed (6-1). Increase Dressing Feed by manually pushing the Dresser harder against the Grinding Wheel (23).
3. It is important that the whole width of the Grinding Wheel is dressed off to the same diameter.



6-1	Manual Feed Handle
6-2	Cutting Feed Handle
23	Grinding Wheel
23-1	Grinding Wheel Cover
23-2	Dresser Assembly
23-3	Diamond Dresser

4. It might become necessary to dress the grinding wheel several times during the grinding operation. Make sure that the dressing is done when the grinding wheel does NOT contact the seat surface of valve spindle or bottom piece.



WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' MAINTENANCE

Feed shafts should be cleaned and greased after each use of the machine, using such grease as Eneos Power knock WB2(JIS Common Use No.1) or equivalent.

Bare metal parts should be kept oiled or coated.

Check condition of Shock Absorbers.

Periodically check the electric motors and if the brushes are worn out, immediately replace with new parts.

The electric motors are fabricated from a sturdy design, however if structural damage occurs it is recommended that the complete motor be replaced, this being the most economical remedy.

Electric Grinding Motors, in general, are not to be continuously run for more than 30 minutes at a time.

The life time of the electric motors is expected to be at least 1000 running hours as long as the motors are operated properly, avoiding over-load and over-heat.

Adjust the tension of the Timing Belts as required.



WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' REPLACEMENT OF TIMING BELT

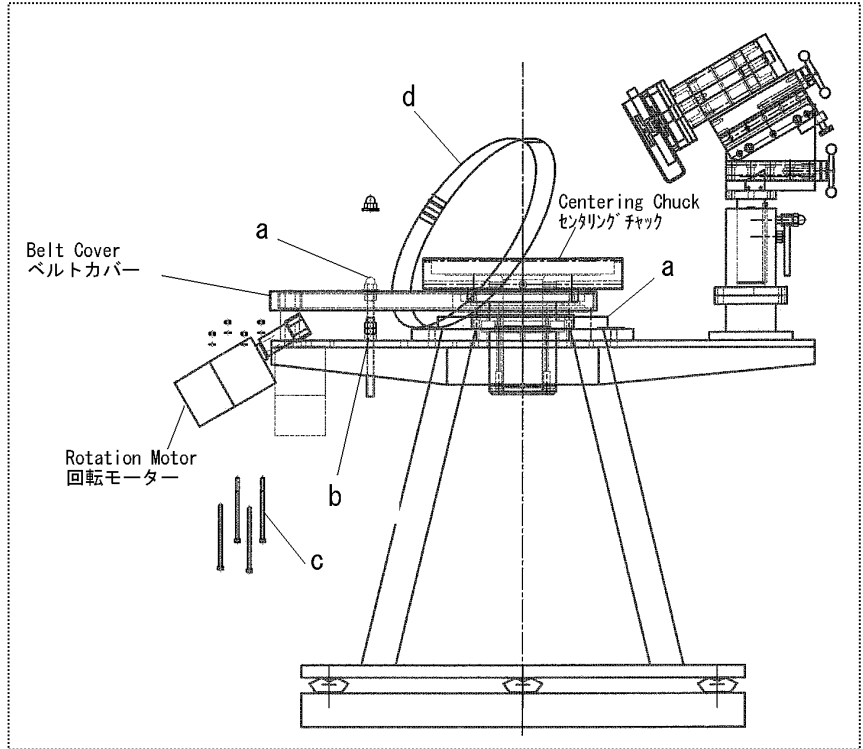
TIMING BELT FOR ROTATION GEAR

① Loosen the securing bolts for the Belt Cover (three places). See (a) on the sketch.

② Unlock the nuts and screw out the Stud Bolt for the Belt Cover at the Rotation Motor side. See (b) on the sketch.

③ Unscrew the securing bolts for the Drive Gear and remove the Gear. Ensure not to pull on the electric cables when placing the Drive Gear on the floor. See (c) on the sketch.

④ Remove the broken Timing Belt. Pull a new Timing Belt over the Centering Chuck and push it under the Belt Cover. See (d) on the sketch.



⑤ Fix it in place and apply tension to the timing belt by pulling the Drive Gear and then securing the Drive Gear with the four securing bolts. See (c) on the sketch.

⑥ Refit the Belt Cover. See (a)-(b) on the sketch.

⑦ Check clearance between Belt Cover and Timing Belt. Confirm tension of the Timing Belt.

TIMING BELT FOR GRINDING SPINDLE

① Unscrew the two Wheel Hub Nuts, it may be necessary to remove the end cover of the Grinding Spindle and hold the shaft preventing it from rotating when removing the inner Wheel Hub Nut.

② Remove the Belt Cover.

③ Unlock the set bolts for the Pulleys and pull out the Pulleys simultaneously.

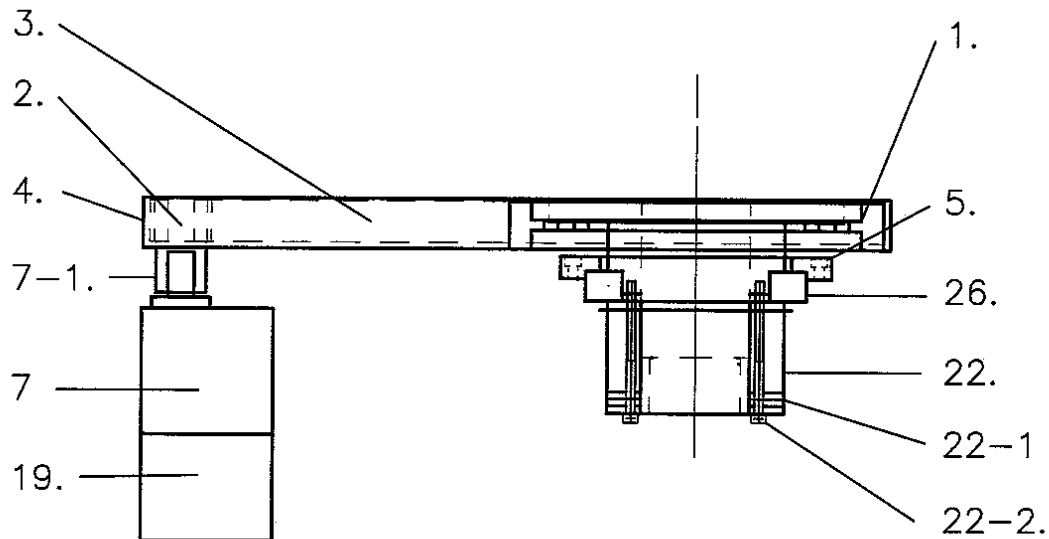
④ Fit the new Timing Belt to the Pulleys and refit the Pulleys simultaneously onto the shafts.

This procedure will ensure the proper tension is applied to the Timing Belt.

⑤ Refit Belt Cover and Wheel Hub Nuts.



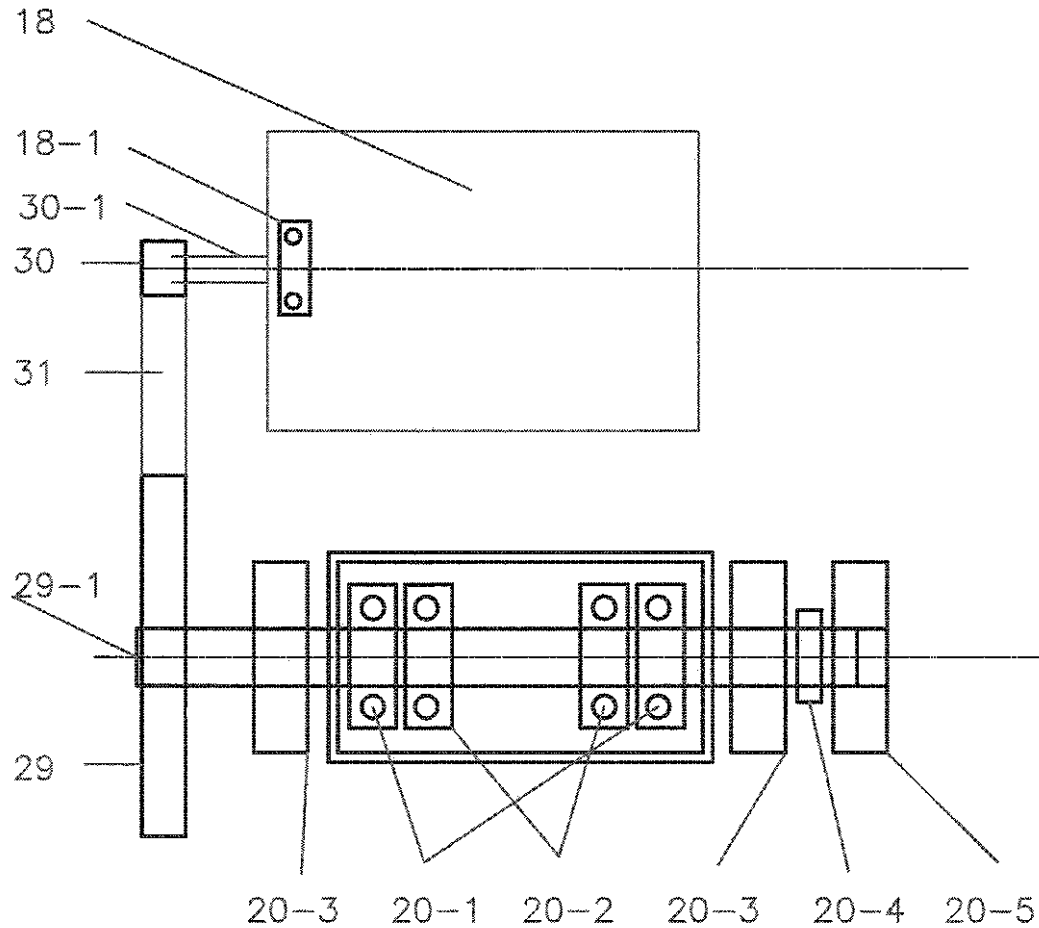
WAKEFIELD EXHAUST VALVE GRINDER **TYPE 'SEG'** **Rotation Gear Parts**



No.	Item	Q'ty
1.	Synchro Pulley L Rotation	1
2.	Synchro Pulley S Rotation	1
3.	Timing Belt Rotation	1
4.	Belt Cover	1
5.	Bearing Cover	1
7.	Gear Box complete	1
7-1.	Shaft Adaptor	1
19.	Driving Motor complete	1
22.	Securing Chuck	1
22-1.	Securing Chuck Centering Bolts	3
22-2.	Securing Tie Bolts	4
26.	Bearing (Cross Roller Ring)	1



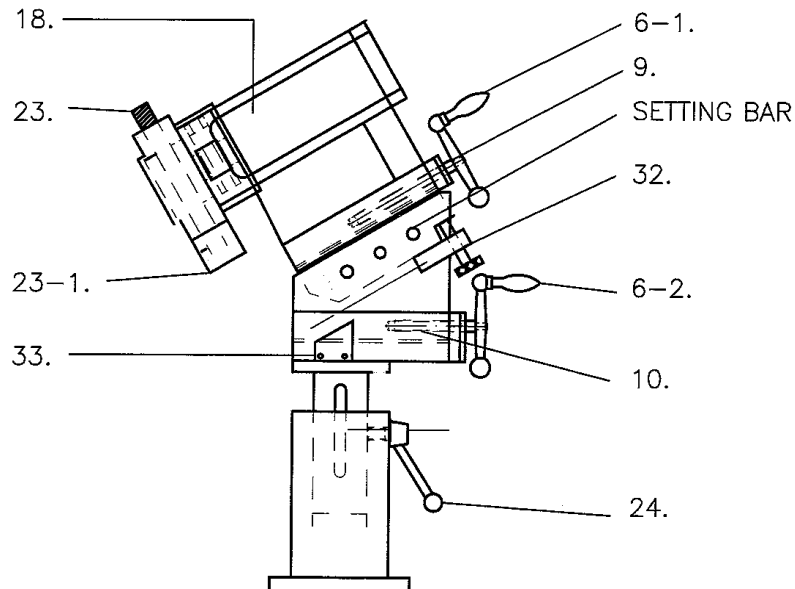
WAKEFIELD EXHAUST VALVE GRINDER **TYPE 'SEG'** **Grinding Spindle Parts**



No.	Item	Q'ty
18	Grinding Motor complete	1
18-1.	Bearing	1
20-1.	Bearing	2
20-2.	Bearing	2
20-3.	Bearing Holder	2
20-4.	Bearing Nut	1
20-5.	End Cover	1
29.	Synchro Pulley L Grinding	1
29-1.	Shaft	1
30.	Synchro Pulley S Grinding	1
30-1	Grinding Motor Shaft	1
31.	Timing Belt for Grinding	1



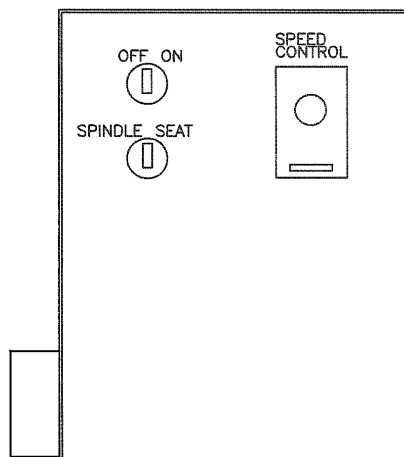
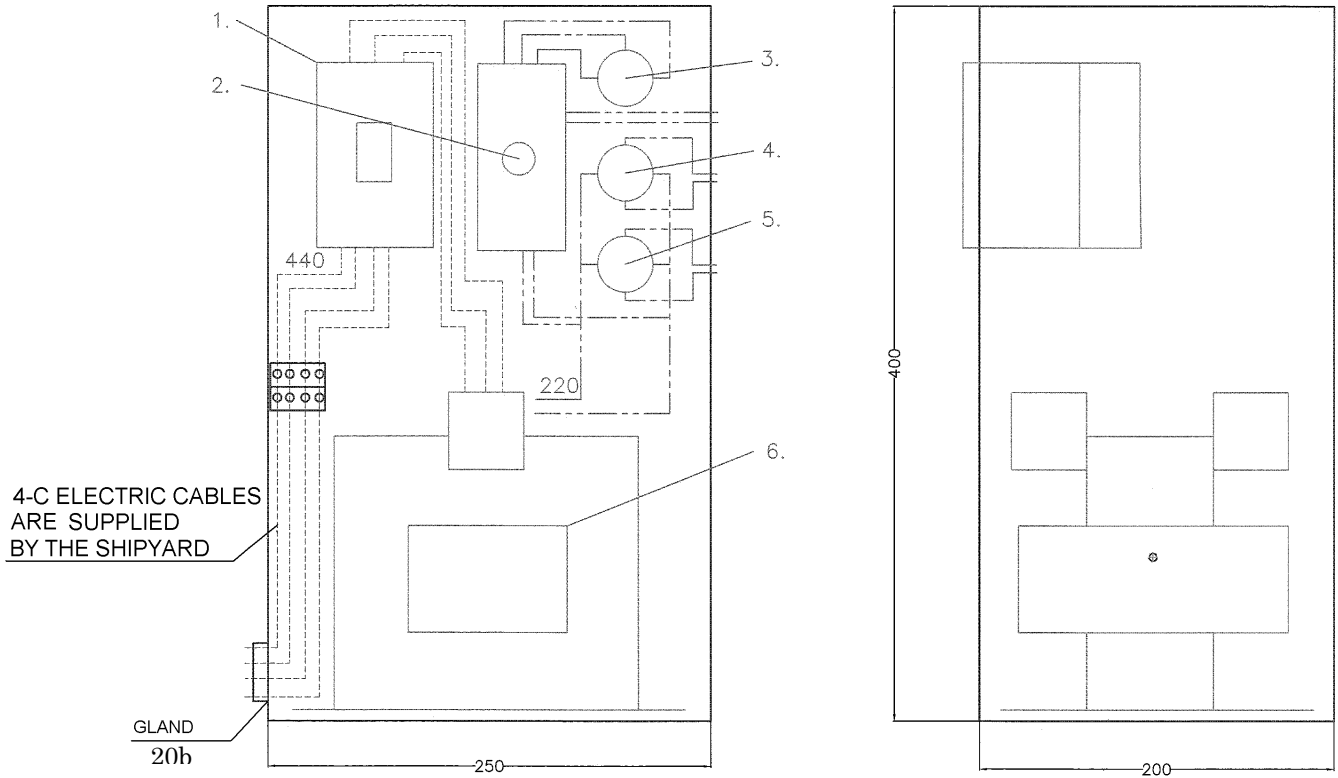
WAKEFIELD EXHAUST VALVE GRINDER **TYPE 'SEG'** **Angle Feed Assembly Parts**



No.	Item	Q'ty
6-1.	Grinding Feed Handle	1
6-2.	Cutting Feed Handle	1
9.	Angle Feed Slide	1
10.	Cutting Feed Slide	1
18.	Grinding Motor complete	1
23.	Grinding Wheel	1
23-1.	Grinding Wheel Cover	1
24.	Lock Grip	1
32.	Angle Adjustment Device	1
33.	Angle Indication Plate	1



WAKEFIELD EXHAUST VALVE GRINDER TYPE 'SEG' 440V AC 3Ph Power Supply Control Box

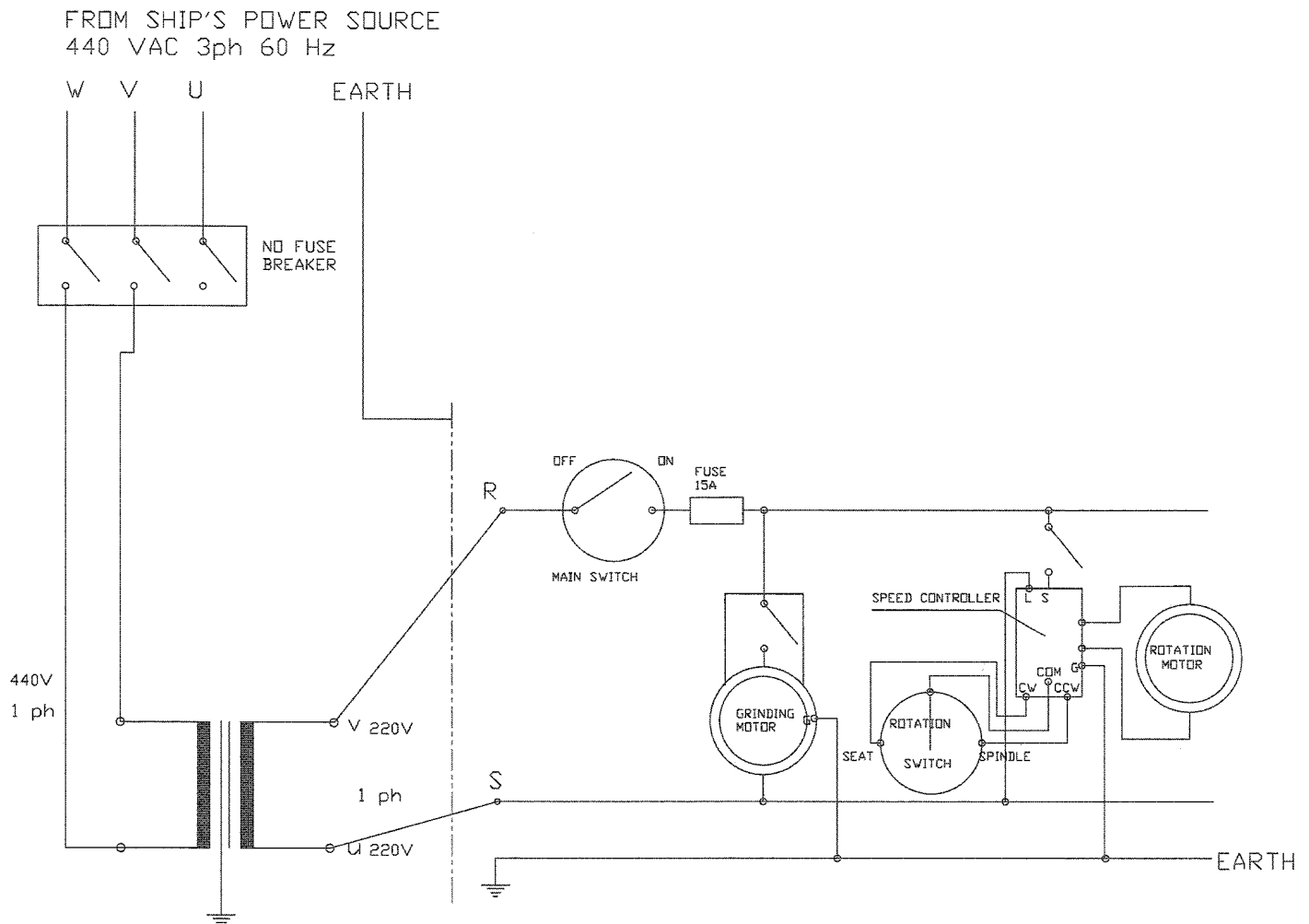


1. No- Fuse Breaker
2. Speed Controller
3. CW/CCW Rotation Knob
4. Grinding Motor On/Off
5. Rotation Motor On/Off
6. Transformer 440-220V, 2kVA

Front Panel



WAKEFIELD EXHAUST VALVE GRINDER **TYPE 'SEG'** **440V AC 3 Ph Power Supply Electric Diagram**



Grinding Machine's currency requirement for 440 VAC 3ph power supply.

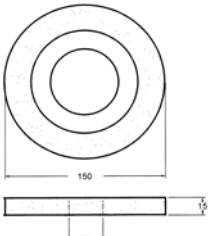
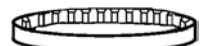
Starting moment:	4.5 Amps
Idling:	2.2 Amps
Working:	3.1 Amps



WAKEFIELD EXHAUST VALVE GRINDER

TYPE 'SEG'

SPARE PARTS

SPARE PARTS LIST FOR EXHAUST VALVE GRINDER "SEG"						
No.	NAME	SKETCH	WORKING	SPARE	WEIGHT	REMARKS
1	Grinding Wheel		1	1	0.4 kgs	P/N. KWM-709 For Valve Seat & Spindle 150 × 15 × 32mm
2	Timing Belt For Rotation Motor		1	1	0.1 kg	P/N. 510L-100
3	Timing Belt For Grinding Motor		1	1	0.02 kgs	P/N. XL170
4	Fuse		1	1	0.005 kgs	15A

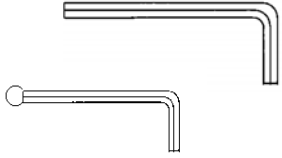
Note: Only original Wakefield parts and Grinding Wheel should be used.



WAKEFIELD EXHAUST VALVE GRINDER

TYPE 'SEG'

TOOL LIST

TOOL LIST FOR EXHAUST VALVE GRINDER TYPE "SEG"					
No.	NAME	SKETCH	Q'ty	weight	REMARKS
1	Hexagon Wrench		Total 4 pcs	0.12 kgs	4mm × 1pc 5mm ball end long × 1pc 6mm × 1pc 8mm × 1pc
2	Grinding Wheel Dresser		1	0.05 kgs	ND-1A6A Φ 11x30
3	Wheel Hub Tightening Bar		2	0.15 kgs	Φ 6mmx100mm
4	Magnetic gauge with dial gauge		1		

Note: Only original Wakefield parts and Grinding Wheel should be used.



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WHEN ORDERING WAKEFIELD HONING AND GRINDING MACHINES, SUITABLE HONING STONES AND GRINDING STONES AND/OR OTHER COMPONENTS OR SERVICES PLEASE CONTACT OUR WORLD-WIDE AGENTS OR OUR HEAD OFFICE IN JAPAN.

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